

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE DEPARTMENT

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Artificial Intelligence
Assignment 3

Semester: 6th

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Batch: 2022F

Max Marks: 10

Section: A

S. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
CLO_3	Apply Advance artificial intelligence techniques to identify problems and provide solutions using AI also discuss the ethical implications of the solutions.	PLO_9 (Ethics)	C3 (Applying)

Q1) Explain the basic concept of a Decision Tree. What are its main components and how is it used in classification tasks?

Basic Concept of a Decision Tree:

A **Decision Tree** is a flowchart-like tree structure used for decision-making and predictive modeling. It splits data into subsets based on feature values, helping to arrive at a decision or prediction.

Main Components of a Decision Tree:

- Root Node:**
 - The top-most node representing the entire dataset.
 - It starts the decision-making process based on the most significant feature.
- Internal (Decision) Nodes:**
 - Represent features or attributes used for splitting.
 - Each internal node asks a question based on a feature (e.g., "Is age > 30?").
- Branches:**
 - Represent the outcome of the test at each node.
 - Each branch leads to another decision node or a leaf node.
- Leaf (Terminal) Nodes:**
 - Represent the final outcome (class label in classification).
 - No further splitting occurs.

How It's Used in Classification Tasks:

- Training:**
 - The algorithm selects features and thresholds to split the dataset.
 - Splitting aims to maximize **information gain** or minimize **impurity** (e.g., Gini Index, Entropy).
 - This continues recursively to form a tree.
- Prediction:**
 - A new instance is passed through the tree.
 - At each node, the relevant feature condition is checked.
 - The instance follows the branch that satisfies the condition until it reaches a leaf node.
 - The class label at the leaf is the prediction.

Q2) Describe the purpose of attribute selection measures in Decision Trees. Compare Information Gain, Gain Ratio, and Gini Index with suitable examples.

Purpose of Attribute Selection Measures in Decision Trees:

Attribute selection measures help the decision tree algorithm decide **which attribute to split on** at each node. The goal is to choose the attribute that best separates the data into distinct classes — i.e., provides the most useful information for classification.

1. Information Gain (IG):

- **Definition:** Measures the reduction in entropy (uncertainty) after a dataset is split on an attribute.
- **Used in:** ID3 algorithm.
- **Formula:** $IG(S,A) = Entropy(S) - \sum (|S_v|/|S| \cdot Entropy(S_v))$
- **Example:**

Suppose we have a dataset S with 10 samples (6 Yes, 4 No). Splitting on attribute "Weather":

- $Entropy(S) = 0.971$
- After splitting on "Weather", new entropy = 0.693
- $\rightarrow \text{Information Gain} = 0.971 - 0.693 = \mathbf{0.278}$

2. Gain Ratio (GR):

- **Definition:** A modification of Information Gain that reduces its bias toward attributes with many distinct values.
- **Used in:** C4.5 algorithm.
- **Formula:** $GR = \text{Information Gain} / \text{Split Information}$
- **Split Information:** Measures how broadly and evenly the attribute splits the data.
- **Example:**
If "Student ID" has a high IG (because it's unique for each instance), its Split Info would also be high, making the GR low — and **discouraging the split**.

3. Gini Index:

- **Definition:** Measures the impurity or probability of incorrect classification of a randomly chosen element.
- **Used in:** CART (Classification and Regression Trees).
- **Formula:** $Gini(S) = 1 - \sum (p_i)^2$
where p_i is the proportion of class i.
- **Example:**
For a node with 4 Yes and 6 No:
 $Gini = 1 - (0.42 + 0.62) = 1 - (0.16 + 0.36) = 0.48$

Comparison Table:

Measure	Main Idea	Bias	Used In	Best For
Information Gain	Reduces entropy	Biased to many values	ID3	Clear class separation
Gain Ratio	Adjusts IG by split info	Less biased	C4.5	Attributes with many values
Gini Index	Measures impurity	Less biased	CART	Binary classification, speed

Q3)Describe the concept of Attribute Selection Measures in Decision Trees. Why are they important?

Concept of Attribute Selection Measures in Decision Trees:

Attribute Selection Measures are criteria used to decide which attribute to split on at each step in the construction of a decision tree. Their purpose is to determine the best attribute that results in the most effective split of the data into subsets that are as pure (homogeneous) as possible.

Why Are They Important?

1. **Improve Classification Accuracy:**
 - Good attribute selection leads to better splits, which help classify unseen data more accurately.
2. **Reduce Tree Complexity:**
 - Proper measures help avoid unnecessary splits and overfitting, resulting in simpler, more general trees.
3. **Increase Learning Efficiency:**

- Selecting the right attribute early speeds up the decision-making process during both training and prediction.
4. **Handle Irrelevant/Redundant Data:**
- Attribute selection helps ignore irrelevant features that do not contribute to decision making.

Common Attribute Selection Measures:

1. **Information Gain** (used in ID3)
 - Chooses attributes that reduce entropy the most.
2. **Gain Ratio** (used in C4.5)
 - Adjusts information gain by the diversity of the attribute.
3. **Gini Index** (used in CART)
 - Measures impurity to select the attribute that produces the purest children.

Q4) A school wants to predict whether a student will pass an exam based on study hours, attendance, and previous grades. Describe how a decision tree could be used and what method might be the best for this case?

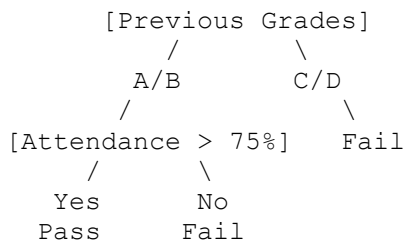
Using a Decision Tree for Predicting Student Exam Results:

A **Decision Tree** can be used to classify whether a student will **pass** or **fail** based on attributes like:

- **Study Hours** (e.g., > 3 hours/day)
- **Attendance** (e.g., > 75%)
- **Previous Grades** (e.g., A, B, C)

How the Decision Tree Works in This Case:

1. **Data Collection:**
 - Collect past student records with features: Study Hours, Attendance, Previous Grades, and target label: Pass or Fail.
2. **Tree Construction (Training):**
 - The algorithm evaluates each attribute using an **attribute selection measure** (like Information Gain, Gain Ratio, or Gini Index).
 - It selects the **best attribute** to split the data at each node.
3. **Example Tree Structure:**



Prediction:

- For a new student, the tree asks questions (nodes) like:
 - What are the previous grades?
 - Is attendance above 75%?
 - Are study hours more than 3/day?
- Based on answers, it reaches a leaf node (Pass/Fail).

Best Method for This Case:

- **Recommended Algorithm: C4.5** using **Gain Ratio**
 - Because:
 - Handles both categorical (Grades) and continuous (Hours, Attendance) data.
 - Gain Ratio avoids bias toward attributes with many unique values (like raw grades).
 - More robust with noisy or real-world educational data.

Q5) In a clinical decision-support system, a physician analyzes patient symptoms such as body temperature (categorized as low, moderate, high), presence of cough, and sore throat to diagnose influenza. Explain how advanced selection measures like Information Gain, Gain Ratio, or Gini Index would influence the construction and accuracy of the resulting decision tree. Which attribute would likely be chosen as the root node, and why?

Using Attribute Selection Measures in Clinical Decision-Support Systems:

A physician wants to diagnose **influenza** based on patient symptoms:

- **Body Temperature:** low, moderate, high (categorical)
- **Presence of Cough:** yes/no (binary)
- **Sore Throat:** yes/no (binary)

How Advanced Attribute Selection Measures Influence Tree Construction & Accuracy:

1. **Information Gain (IG):**
 - Measures how much uncertainty (entropy) is reduced by splitting on an attribute.
 - It tends to favor attributes with **many distinct values** (e.g., temperature with 3 categories).
 - If temperature clearly separates influenza vs. no influenza cases, IG for temperature will be high.
2. **Gain Ratio (GR):**
 - Adjusts IG by considering how broadly the attribute splits data.
 - Penalizes attributes with many unique values to avoid bias.
 - Helps choose an attribute that is both informative and balanced in splitting.
3. **Gini Index:**
 - Measures impurity (probability of misclassification).
 - Picks the attribute that minimizes impurity after the split.
 - Often faster and effective for categorical/binary attributes.

Likely Root Node Choice:

- **Body Temperature** is likely to be chosen as the root node because:
 - It has **3 distinct categories** which may strongly correlate with influenza (e.g., high temp → higher chance).
 - It likely provides the **highest information gain** or the best impurity reduction.
- However, if temperature splits unevenly or is biased, **Gain Ratio** might favor a binary attribute like **Cough**, if it better balances the split.

Impact on Accuracy:

- Choosing the right root attribute influences **tree depth** and **classification accuracy**.
- Using **Gain Ratio** or **Gini Index** reduces overfitting by avoiding biased splits.
- This results in a **more generalizable** and **accurate** clinical decision-support model.

Summary:

Attribute	Why Likely Root?	Measure Impact
Body Temperature	Strong predictor, multi-category split	High IG but adjusted by GR
Presence of Cough	Simple binary, may have balanced split	May be preferred by Gini or GR
Sore Throat	Binary, less predictive in many cases	Lower IG/Gini compared to others

